

Incorrect-Answer Audit

Every mismatch from the Qwen3-VL Plus Gear Assemblies base-package evaluation

Field	Value
Model	qwen3-vl-plus
Evaluation date	2026-08-11
Repository commit	360772e84d93f63010c806fc0faf21654bb8c4fc
Scope	33 top-level base packages; hidden content excluded

Purpose: document methodology, exact-answer performance, question-type error rates, token use, estimated list-price cost, and per-item failures.

Prepared from retained provider responses, blind grading outputs, and independent reconciliation checks.

Audit overview

This companion report enumerates all 28 incorrect answers. Every record includes the original question, model prediction, correct answer, signed numeric error where applicable, and a concise descriptive diagnosis. Signed error equals prediction minus correct answer. Negative values are underestimates; positive values are overestimates. The gear-type item is categorical.

The error set contains 27 numeric mismatches and one discrete mismatch. Among numeric errors, 21 were underestimates and 6 were overestimates. Transmission ratios contributed 10 errors, tooth counts 13, component counts 4, and gear type 1. Axis angle contributed no error.

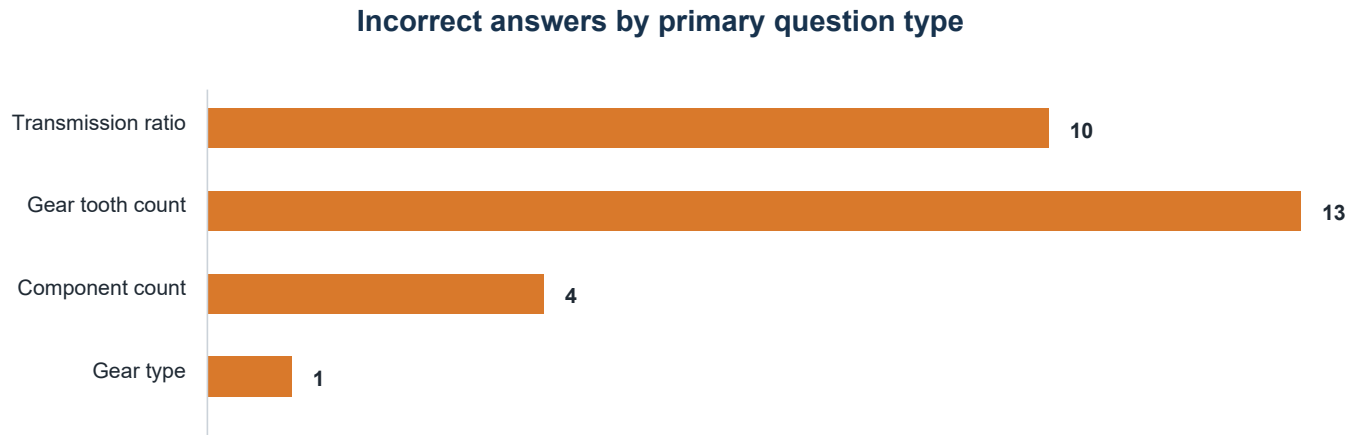


Figure 1. Incorrect-answer counts by primary question type.

Transmission ratio

Package	Subtype	Question	Predicted	Correct	Signed error	Diagnostic
Model01_QA 01	Worm train ratio	what is the transmission ratio of this gear train? Assume the worm is a single-start.	10	30	-20.00; -66.7%	The transmission ratio was underestimated by 20 (66.7% of the correct value).
Model02_QA 01	Other train or gearbox ratio	what is the transmission ratio of the gear train?	1	16	-15.00; -93.8%	The transmission ratio was underestimated by 15 (93.8% of the correct value).
Model03_QA 01	Planetary ratio	what is the transmission ratio of this planetary gear train? Assume the sun gear is the input, ring gear is fixed. And carrier is the output.	3	6	-3.00; -50.0%	The transmission ratio was underestimated by 3 (50.0% of the correct value).
Model04_QA 01	Planetary ratio	what is the transmission ratio of this planetary gear? Assume sun input, ring fixed, carrier output. Round up to the nearest integer.	4	5	-1.00; -20.0%	The transmission ratio was underestimated by 1 (20.0% of the correct value).
Model05_QA 03	Other train or gearbox ratio	what is the transmission ratio of this gear train?	5	4.65	+0.35; +7.5%	The transmission ratio was overestimated by 0.35 (7.5% of the correct value).
Model06_QA 04	Other train or gearbox ratio	What is the transmission ratio of this gear box?	3	2	+1.00; +50.0%	The transmission ratio was overestimated by 1 (50.0% of the correct value).
Model07_QA 03	Other train or gearbox ratio	What is the transmission ratio of this gear train?	1	12	-11.00; -91.7%	The transmission ratio was underestimated by 11 (91.7% of the correct value).
Model08_QA 03	Other train or gearbox ratio	What the transmission ratio of this gear box? Round to the first decimal place.	2.0	3.6	-1.60; -44.4%	The transmission ratio was underestimated by 1.6 (44.4% of the correct value).
Model09_QA 02	Other train or gearbox ratio	What is the transmission ratio of the gear box? Round to the first decimal place.	3.0	5.4	-2.40; -44.4%	The transmission ratio was underestimated by 2.4 (44.4% of the correct value).
Model10_QA 01	Other train or gearbox ratio	what is the transmission ratio of this gear train?	10	32	-22.00; -68.8%	The transmission ratio was underestimated by 22 (68.8% of the correct value).

Gear tooth count

Package	Subtype	Question	Predicted	Correct	Signed error	Diagnostic
Model03_QA 02	Ring gear teeth	What is the tooth number of the ring gear?	72	80	-8.00; -10.0%	The tooth count was underestimated by 8 (10.0% of the correct value).
Model03_QA 03	Sun gear teeth	What is tooth number of the sun gear?	24	20	+4.00; +20.0%	The tooth count was overestimated by 4 (20.0% of the correct value).
Model04_QA 02	Ring gear teeth	What is the tooth number of the ring gear?	48	65	-17.00; -26.2%	The tooth count was underestimated by 17 (26.2% of the correct value).
Model04_QA 03	Sun gear teeth	What is tooth number of the sun gear?	24	16	+8.00; +50.0%	The tooth count was overestimated by 8 (50.0% of the correct value).
Model04_QA 04	Planet gear teeth	What is the tooth number of planet gear?	24	34	-10.00; -29.4%	The tooth count was underestimated by 10 (29.4% of the correct value).
Model05_QA 01	Smaller gear teeth	What is the tooth number of the smaller gear?	12	20	-8.00; -40.0%	The tooth count was underestimated by 8 (40.0% of the correct value).
Model05_QA 02	Larger gear teeth	What the tooth number of the larger gear.	60	93	-33.00; -35.5%	The tooth count was underestimated by 33 (35.5% of the correct value).
Model06_QA 02	Smaller gear teeth	What the tooth number of the smaller gear?	12	25	-13.00; -52.0%	The tooth count was underestimated by 13 (52.0% of the correct value).
Model06_QA 03	Larger gear teeth	What is the tooth number of the larger gear?	24	50	-26.00; -52.0%	The tooth count was underestimated by 26 (52.0% of the correct value).
Model08_QA 01	Smaller gear teeth	what the tooth number of the smaller gear?	12	29	-17.00; -58.6%	The tooth count was underestimated by 17 (58.6% of the correct value).
Model08_QA 02	Larger gear teeth	what the tooth number of the larger gear?	24	105	-81.00; -77.1%	The tooth count was underestimated by 81 (77.1% of the correct value).
Model09_QA 01	Sun gear teeth	What is the tooth number of the sun gear?	24	17	+7.00; +41.2%	The tooth count was overestimated by 7 (41.2% of the correct value).
Model09_QA 03	Ring gear teeth	What is the tooth number of the ring gear?	72	75	-3.00; -4.0%	The tooth count was underestimated by 3 (4.0% of the correct value).

Component count

Package	Subtype	Question	Predicted	Correct	Signed error	Diagnostic
Model02_QA02	Assembly inventory count	How many support brackets are there in this assembly?	3	6	-3.00; -50.0%	The component count was underestimated by 3 (50.0% of the correct value).
Model06_QA01	Assembly inventory count	How many bearings are there is in this gearbox?	4	2	+2.00; +100.0%	The component count was overestimated by 2 (100.0% of the correct value).
Model07_QA01	Assembly inventory count	How many gears are there in this gear box?	3	4	-1.00; -25.0%	The component count was underestimated by 1 (25.0% of the correct value).
Model07_QA02	Assembly inventory count	How many shafts are there in this gear box?	2	3	-1.00; -33.3%	The component count was underestimated by 1 (33.3% of the correct value).

Gear type

Package	Subtype	Question	Predicted	Correct	Signed error	Diagnostic
Model08_QA04	Gear-shape classification	What is the type of the gear shape? Spur, Helical, Bevel or Double Helical or others.	Bevel	Helical	categorical	Classified as Bevel instead of Helical.

Correct-answer control cases

These five items were correct and account for the remainder of the 33-question denominator.

Package	Type	Question	Correct answer	Prediction
Model01_QA02	Axis angle	What is the angle between the axes of the worm and the worm wheel?	90	90
Model02_QA03	Component count	How many distinct parts are there in this assembly?	5	5
Model03_QA04	Component count	How many planet gears are there in this planetary gear train?	3	3
Model04_QA05	Component count	How many planet gears are there in this planetary gear train?	3	3
Model09_QA04	Component count	How many planet gears are there?	4	4

Error-pattern synthesis

All transmission-ratio and tooth-count questions were incorrect. Ratio predictions were usually too low, with especially large relative misses for Model02_QA01, Model07_QA03, and Model10_QA01. Tooth counts also skewed low. The largest absolute miss occurred on Model08_QA02, where 24 was predicted instead of 105. Several small sun-gear targets were overestimated.

Component counts separated into two patterns. All planet-gear quantity items were correct, while support brackets, bearings, total gears, and shafts were not. The single discrete error labeled a helical gear as bevel.

Interpretation limits

The diagnostics describe observed answer differences and do not establish internal model mechanisms. Direct-answer generation suppressed reasoning, so the audit cannot distinguish perception, cross-view correspondence, tooth enumeration, kinematic setup, arithmetic, and output-selection errors. Hidden packages remained outside scope.

References

[1] Alibaba Cloud. Qwen-VL compatible with OpenAI. Accessed Aug. 11, 2026. <https://www.alibabacloud.com/help/en/model-studio/qwen-vl-compatible-with-openai>

[2] axioforgeai. gear-assemblies-qa. GitHub commit 360772e84d93f63010c806fc0faf21654bb8c4fc. <https://github.com/axioforgeai/gear-assemblies-qa/tree/360772e84d93f63010c806fc0faf21654bb8c4fc>